

B.Sc (HONS.) IN CSE, FIRST YEAR FIRST SEMESTER EXAMINATION, 2023

STRUCTURED PROGRAMMING LANGUAGE

[According to the New Syllabus]

Subject Code : CSE-510201

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| 1. (a) What do you mean by structured programming language? Explain different control structures used in structured programming language. | 1+4=5 |
| (b) What do you mean by preprocessor directive? Why and when do we use # include and # define directives? | 1+4=5 |
| (c) Draw a flowchart to determine the roots of a quadratic equation, $ax^2 + bx + c = 0$. | 5 |
| (d) Write down the differences between increment and decrement operators. | 5 |
| 2. (a) What do you mean by variable? How can you declare and initialize a variable using C programming. | 4 |
| (b) Write down the steps of executing a C program. | 5 |
| (c) What are keywords and identifiers? Define with proper examples. | 5 |
| (d) Write a program in C to calculate the area and circumference of a circle. | 6 |
| 3. (a) Explain call by value and call by reference with example. | 5 |
| (b) Define symbolic constant. How does variable and symbolic constant differ? Give example. | 1+4=5 |
| (c) Describe logical and relational operators with example. | 5 |
| (d) Write a C program to input an integer number and display the sum of its digits. | 5 |
| 4. (a) Define loop and looping process. Explain entry controlled and exit controlled loops. | 6 |
| (b) What are the differences between while and do while loop? | 3 |
| (c) Write down the general form and flow chart of the following statement : | 6 |
| (i) else if ladder | |
| (ii) SWITCH | |
| (iii) FOR | |
| (d) Write a C program to find first n Fibonacci numbers. | 5 |

[Please turn over

- | | Marks |
|---|-------|
| 5. (a) What do you mean by linear array? Describe the declaration and initialization of a linear array. | 1+4=5 |
| (b) Define pointer. Write a C program to read an array and display its element in reverse order using pointer. | 1+5=6 |
| (c) What is string? Describe the following string handling functions :
(i) gets () and puts ()
(ii) strcpy () and strcmp () | 1+4=5 |
| (d) Determine the output of the following program segment :
<pre> int x = 2 , y = 3 ; x = x + y; y = x - y; x = x - y; printf("x is %d\n y is %d\n", x, y); </pre> | 4 |
| 6. (a) What is file? Describe different file opening modes. | 1+4=5 |
| (b) What do you mean by dynamic memory allocation? What are the advantages of using linked list over array? | 3 |
| (c) Write down the differences between printf () and fprintf (). | 4 |
| (d) Write a C program to read data from keyboard write it to a file called INPUT, again read the same data from the INPUT file and display it on the screen. | 6 |

B.Sc (HONS.) IN CSE, FIRST YEAR FIRST SEMESTER EXAMINATION, 2023
ELECTRICAL AND ELECTRONIC CIRCUIT

[According to the New Syllabus]

Subject Code : 510203

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | | Marks |
|---|---|-------|
| ① | (a) State and explain voltage divider rule. | 4 |
| | (b) State and explain maximum power transfer theorem. | 5 |
| | (c) Find the current through the 10Ω resistor of the network in fig 1(c) : | 6 |

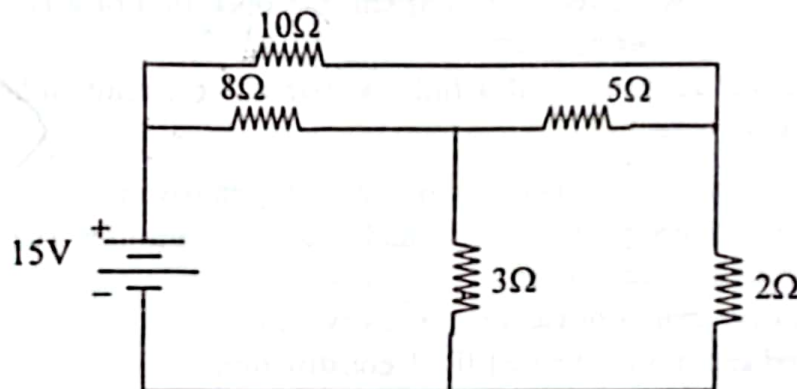


Fig : 1 (c)

- | | | |
|---|--|---|
| | (d) Which factors of a conductor depends on its resistance? Explain. | 5 |
| ② | (a) State and explain Thevenin's theorem. | 5 |
| | (b) Find the Thevenin equivalent circuit for the network in fig 2(b) : | 5 |

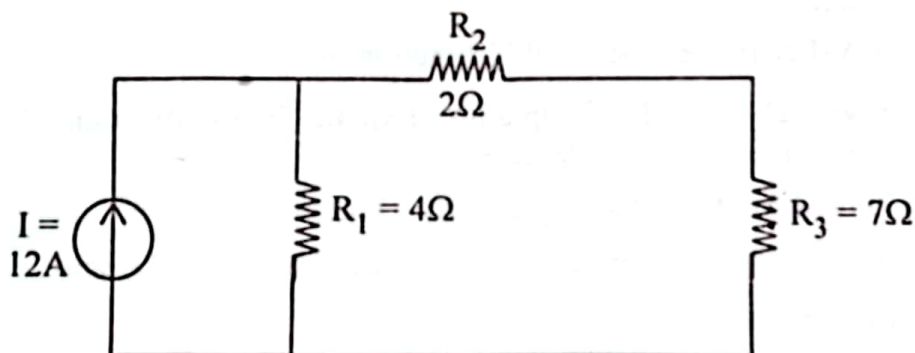


Fig : 2 (b)

- | | | |
|--|--|---|
| | (c) Distinguish between series and parallel circuit. | 5 |
| | (d) Discuss the effect of temperature on resistance. | 5 |

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- ③ (a) State and explain Norton's theorem.
 (b) Find the Norton equivalent circuit for the network in fig 3(b) :

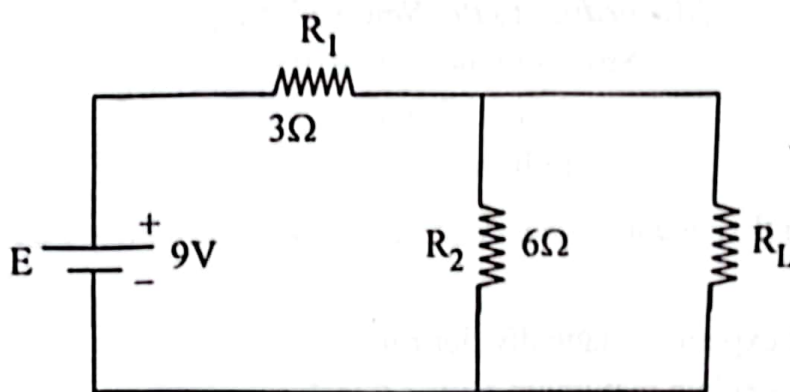


Fig : 3 (b)

- (c) What is a half wave rectifier? Explain the operation of a H.W rectifier with proper diagram. 5
- (d) Mention the advantages of a full wave rectifier circuit rather than half wave rectifier. 3
4. (a) Derive the relationship between α and β of a transistor. 5
- (b) Compare Common Base (CB) and Common Emitter (CE) configuration of transistor. 5
- (c) Explain how transistor can be used as switch. 5
- (d) Draw and explain N-channel JFET construction. 5
- ⑤ (a) Define pinch-off voltage, DC load line and operating point of transistor. 3+2=5
- (b) Explain working principle of enhancement type MOSFET (N-channel). 5
- (c) What is semiconductor? Describe p-type and n-type semiconductors. 1+4=5
- (d) Explain V-I characteristics of P-N Junction diode. 5
6. (a) Define virtual ground of Op-amp. Explain block diagram representation of a typical Op-amp. 2+6=8
- (b) Explain the operation of an Op-amp as integrator. 6
- (c) In an inverting amplifier let $R_F = 100 \text{ k}\Omega$, $R_1 = 10 \text{ k}\Omega$ and $V_1 = 1 \text{ V}$. Calculate— 6
- Input Current
 - Output Voltage
 - Closed Loop Gain.

**B.Sc (HONS.) IN CSE FIRST YEAR FIRST SEMESTER
EXAMINATION, 2023
CALCULUS**

[According to the New Syllabus]

Subject Code: CSE-510205

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Group—A (Differential Calculus)

- | | Marks |
|---|--------|
| 1. (a) Define domain and range with examples. | 4 |
| (b) Test the continuity and differentiability of $f(x)$ at $x = 0$ and $x = \frac{\pi}{2}$, | 10 |
| $\text{if } f(x) = \begin{cases} 1 & \text{when } -\infty < x < 0 \\ 1 + \sin x & \text{when } 0 \leq x \leq \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & \text{when } \frac{\pi}{2} < x < \infty. \end{cases}$ | |
| (c) Define limit of a function. By (δ, ϵ) definition prove that, | 2+4=6 |
| $\lim_{x \rightarrow 3} \frac{2x^2 - 18}{x - 3} = 12.$ | |
| 2. (a) Find $\frac{dy}{dx}$ (any three): | 4×3=12 |
| (i) $xy = \sin(x + y)$ | |
| (ii) $y = x^x + x^{\frac{1}{x}}$ | |
| (iii) $x^y + y^x = a^b$ | |
| (iv) Differentiate $x^{\sin^{-1}x}$ with respect to $\sin^{-1}x$ | |
| (b) State and prove Leibnitz's theorem. | 8 |
| 3. (a) State and prove Rolle's theorem. | 8 |
| (b) Verify Rolle's theorem for $f(x) = x^2 - 3x + 2$ in the interval $(1, 2)$. | 6 |
| (c) Find the maximum and minimum value of the function
$f(x) = x^5 - 5x^4 + 5x^3 - 1.$ | 6 |

[Please turn over



Group—B (Integral Calculus)

Marks
4×5=20

4. Evaluate (any five):

(i) $\int \sin^4 x dx$

(ii) $\int \sin^5 x \cos^3 x dx$

(iii) $\int \frac{e^{a \sin^{-1} x}}{\sqrt{1-x^2}} dx$

(iv) $\int \frac{dx}{3 \sin x + 2 \cos x + 5}$

(v) $\int \frac{dx}{(x^2-1)\sqrt{x^2+1}}$

(vi) $\int x^2 e^x dx$

5. (a) State and prove fundamental theorem of integral calculus.

8

(b) Using Gamma and Beta function solve:

4

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{3}{2}} \theta \cos^3 \theta d\theta$$

(c) Evaluate (any two):

4×2=8

(i) $\int_0^{\frac{\pi}{2}} \frac{dx}{3+2 \sin x}$

(ii) $\int_0^{16} \frac{x^{\frac{1}{4}}}{1+\sqrt{x}} dx$

(iii) $\int_0^{\infty} \frac{(\tan^{-1} x)^2}{1+x^2} dx$

6. (a) Find the length of the arc of the parabola $y^2 = 16x$ between the vertex and one end of the latus rectum.

6

(b) Show that the area between the parabola $y^2 = 4x$ and the straight line $y = 2x - 4$ is 9.

7

(c) Find the volume of the solid generated by the revolution of the curve $y^2(a+x) = x^2(a-x)$ about x-axis.

7

**B.Sc (HONS.) IN CSE FIRST YEAR, FIRST SEMESTER
EXAMINATION, 2023**

PHYSICS

[According to the New Syllabus]

Subject Code: 510207

Time—3 hours

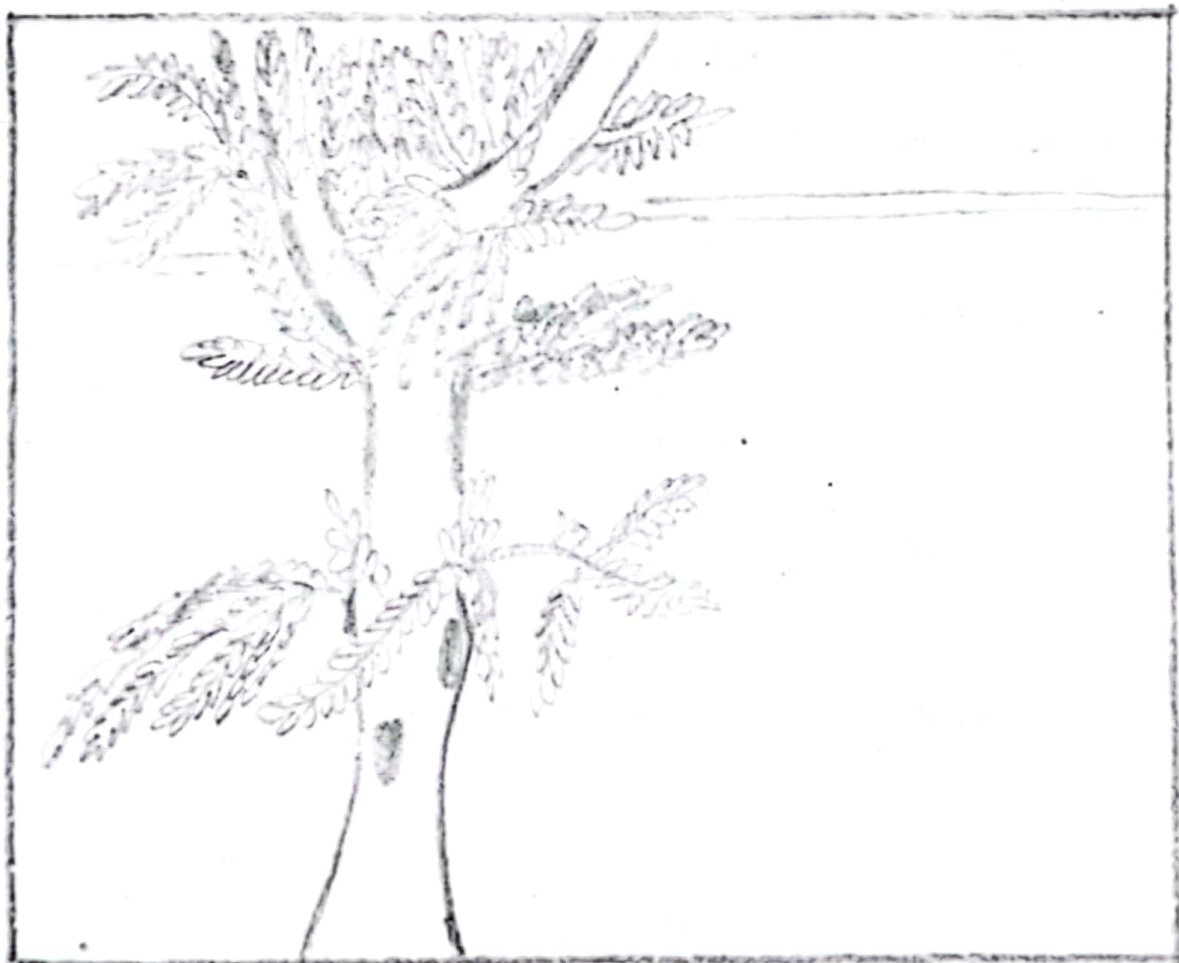
Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| ✓ 1. (a) State and explain Gauss's law. | 2+2=4 |
| (b) Deduce the expression of Gauss's law when a dielectric is present in the electric field. | 6 |
| (c) Briefly explain quantization of charge. | 4 |
| (d) Derive the expression of potential at a distance (r) from an isolated point charge (q). | 6 |
| 2. (a) Draw an R-C circuit and explain transient response of this circuit. | 8 |
| (b) Define solenoid. Find out an expression for magnetic field $\vec{B}$ for a solenoid. | 8 |
| (c) Distinguish between self inductance and mutual inductance. | 4 |
| ✓ 3. (a) State Biot-Sarvart's law. Also express it in vector form. | 6 |
| (b) What is capacitance? Derive an expression for the energy stored in a charged capacitor. | 2+6=8 |
| (c) If three capacitors with capacitances 4, 3 and 2 μF are connected such that the first and second are in series combination and third is in parallel combination with them. What is their equivalent capacitance? | 6 |
| 4. (a) Define the terms current, current density, resistance and resistivity. | 4 |
| (b) State and explain Lenz's law. | 5 |
| (c) Draw and explain the basic operation of the Ammeter. | 5 |
| (d) What is Coloumb's law? Deduce Coloumb's law from Gauss's law. | 6 |

[Please turn over]

5. ✓ (a) Define magnetic field and magnetic field strength. 3
- (b) Describe hysteresis curve of magnetization. 5
- (c) State and explain Faraday's law of electro-magnetic induction. 6
- (d) Distinguish among diamagnetic, paramagnetic and ferromagnetic materials. 6
6. ✓ (a) Define electric dipole and electric dipole moment. 2+2=4
- (b) Find the expressions for intensity at a point on the perpendicular bisector of a dipole. 6
- (c) Explain basic operation of Wheatstone bridge with necessary diagram. 5
- (d) Calculate the amount of torque produced on a current carrying loop placed in a magnetic field. 5



**B.Sc (HONS.) IN CSE FIRST YEAR, FIRST SEMESTER
EXAMINATION, 2023**

ENGLISH

[According to the New Syllabus]

Subject Code: 510209

Time—3 hours

Full marks—80

*[N.B. The figures in the right margin indicate full marks. Answer any four questions.
Each part of a question must be answered sequentially.]*

Marks

✓ 1. Read the following passage and answer the questions below:

An intellectual is one who is an enlightened person. He has to give light to others who are in need of it. In every society we find intellectuals such as Philosophers, Scientists, Scholars, Writers and Critics and they, as enlightened men, have a great responsibility towards society and in a society all cannot be intellectuals. If a time comes when all are intellectuals, it would be a blessed time indeed but at present, at any rate, all are not intellectuals and those who are intellectuals have the great responsibility of guiding others on to the right path. If today we have our civilization and culture. If we have order and security in life and if our life is better than that of our primitive ancestors. It is because the intellectuals, from time to time, have been guiding humanity on the path of felicity and amity. An intellectual should come out of his ivory tower and try to elevate others to his level. This is the theme of Tennyson's famous poem "The Palace of Art". An intellectual has the duty of seeing the truth and teaching it to others. An intellectual contemplates on the eternal laws of the universe to explore the truth. The perception of the truth is almost the same as the perception of beauty and the duty of an intellectual is to see this truth or beauty and to reveal it to others.

(a) (i) Who are the intellectuals according to the writer? 5

(ii) What are the responsibilities of the intellectuals?

(b) What is the main idea and what are the supporting ideas in this passage? 5

(c) Write down the meaning of the following words in English and make your own sentences with them (any five): 5

Universe, eternal, enlightened, intellectual, primitive, perception, theme.

(d) Write a summary of the passage. 5

✓ 2. (a) Frame WH-question from the following sentences: 5

(i) Death is preferable to dishonour.

(ii) He is dismissed for his misconduct.

(iii) He listened to what I said.

(iv) Air raids were taking place every night.

(v) I am very glad to meet you.

[Please turn over

- (b) Use right form of verbs in the following:
- (i) If I see him, I (tell) _____ him.
 - (ii) Where _____ you (meet) him last year?
 - (iii) I thought that he (succeed) _____.
 - (iv) The level of wealth (be) _____ low, our development is slow.
 - (v) I (not meet) _____ the leader he left early.

- (c) Correct the following:
- (i) He contacted with the authority.
 - (ii) He seems to say the truth.
 - (iii) The boy resembles to his father.
 - (iv) I hope I am out of pocket.
 - (v) He is working for two hours.

- (d) Combine each of the following groups of sentences into one sentence:
- (i) Cricket is an English game. This game is very popular in our country. At the moment cricket is the number one game in Bangladesh. Bangladesh is also a test playing country.
 - (ii) Bangladesh is facing the danger of ecological imbalance. The danger has resulted from deforestation. People are cutting down trees.

- ✓ (a) Amplify the following: 10
Empty vessels sound much

- (b) Translate into English: 10
আমরা ইংরেজি ভাষাতে খুব দুর্বল। বর্তমানে ইংরেজি, গণিত আর তথ্য প্রযুক্তিতে ~~দক্ষতা~~ ব্যতীত উন্নতি সম্ভব নহে। বাণিজ্যিক কর্মকাণ্ডে ইংরেজি জানা খুবই জরুরি। গণিত চিন্তাশক্তির বিকাশ ঘটায়। গণিতের পাশাপাশি তথ্য প্রযুক্তি বিষয়ে জ্ঞান অর্জন অত্যাৱশ্যক।

4. Write an essay on any one of the following: 20

- (a) Smart Bangladesh
- (b) Tree Plantation
- (c) An Ideal Student
- (d) Unemployment Problem.

5. (a) Write a dialogue between two friends about favourite personality. 10
(b) Suppose you are a reporter of a national daily. Now write a report on rising of prices of essential commodities in the market. 10

- ✓ 6. Write a paragraph on any one of the following: 10

- (a) (i) Brain Drain
- (ii) Food Adulteration.
- (b) Write an application to the editor of a newspaper to publish an article for modernization the traffic system of Dhaka City. 10

National University, Gazipur

B.Sc. (Hons.) in Computer Science & Engineering
Part-1, 1st Semester Examination-2023

Course Code : 510204

(Electrical and Electronic Circuits Lab)

Total Marks: 40

Time: 3 Hours

Answer any one question of the following-

1. To verify Ohm's law and find out its characteristics
2. To verify Voltage Divider Rule (VDR).
- ~~3. To verify Kirchhoff's Current Law (KCL).~~
4. To verify Kirchhoff's Voltage Law (KVL).
5. To verify Thevenin's theorem.
6. To verify Norton's theorem.
7. To construct RLC series circuit and find out its resonant frequency.
8. To determine the V-I Characteristics of a diode.
9. To determine the V-I Characteristics of a Zener diode.
10. To verify the input characteristics of BJT in Common Emitter Configuration (CE).
11. To verify the output characteristics of BJT in Common Emitter Configuration (CE).
12. To design and construct a switching circuit using BJT.

Marks Distribution:

1. Design	= 10
2. Circuit implementation	= 15
3. Result	= 10
4. Viva-voce	= 05
Total	= 40

Ki

$I_1, I_2, I_3, I_1 + I_3$